Syllabus:
↳ Nested loops
↳ Pattern Printing

FAST-NUCES-CFD
PF-LAB
↳ Quiz taken to evaluate some labs

Quiz 01 – 4th Nov 2025

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Write the outputs.

```
int rows = 4;
for (int i = 1; i <= rows; i++) {
        for (int j = i; j < rows; j++) cout << " ";
        for (int j = 1; j <= 2 * i - 1; j++) cout <<
"*";
        cout << endl;
}
for (int i = rows - 1; i >= 1; i--) {
        for (int j = rows; j > i; j--) cout << " ";
        for (int j = 1; j <= 2 * i - 1; j++) cout <<
"*";
        cout << endl;}
```

```
1 2 3 4
1 _ _ _ * 1       *
2 * *     2     * * *
3         3   * * * * *
4         4 * * * * * * *
3         3   * * * * *
2         2     * * *
1         1       *
0         0 _
          ↑
        cursor
```

```
int rows =6;
for (int i = 0; i < rows; i++) {
        int val = 1;
        for (int j = 0; j <= i; j++) {
                cout << val << " ";
                val = val * (i - j) / (j + 1);
        }
        cout << endl;}
```

```
1
1 0
1 0 0.5
1 0 3
```

```
int rows =4;
for (int i = 1; i <= rows; i++) {
        for (int j = 0; j < rows; j++) {
                cout << (i + j - 1) % rows + 1 << " ";
        }
        cout << endl;}
```

```
1 2 4
2 3 4 1
3 4 1 2
4 1 2 3
```

```
Int rows = 4;
for (int i = 1; i <= rows; i++) {
        for (int j = 1; j <= i; j++) {
                if (j == 1 || j == i || i == rows)
cout << "*";
                else cout << " ";
        }
        cout << endl;}
```

```
X
X X
X   X
X X X X
```